ESABCC Indicator Database — automated refresh report

Eurostat / EEA pull · fact-check before merge

Generated 2026-11-01T12:43:21.297Z by *scripts/esabcc-indicators/refresh-from-sources.mjs*. **196** data points added/updated across **61** indicators; **9** recipe(s) errored. Each value below was pulled directly from the linked Eurostat API / EEA data-viewer endpoint and flagged **afterReport** in the data file. Please confirm each new value against its source before merging.

Indicator id	Status	New points (year = value)	Source (click)	Note
esabcc-o2-pec	**updated**	**2022 = 14644.7, 2023 = 14045.2, 2024 = 13883**	Eurostat nrg_ind_eff · primary energy consumption (PEC_EED) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_ind_eff?format=JSON&geo=EU27_2020&nrg_bal=PEC_EED&unit=MTOE	Mtoe×11.63→TWh.
esabcc-o2-fec	**updated**	**2022 = 10704.7, 2023 = 10388.7, 2024 = 10471.8**	Eurostat nrg_ind_eff · final energy consumption (FEC_EED) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_ind_eff?format=JSON&geo=EU27_2020&nrg_bal=FEC_EED&unit=MTOE	Mtoe×11.63→TWh.
esabcc-e5-electrification	**updated**	**2022 = 22.93, 2023 = 23.08, 2024 = 23.44**	Eurostat nrg_bal_c · electricity ÷ total final energy consumption · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_c?format=JSON&geo=EU27_2020&nrg_bal=FC_E&siec=E7000&unit=GWH	Stored as percent-number to match ECNO duplicate end-use-electrification.
esabcc-i6-industry-electrification	**updated**	**2022 = 33.62, 2023 = 33.59, 2024 = 34.02**	Eurostat nrg_bal_c · industry electricity ÷ industry final energy · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_c?format=JSON&geo=EU27_2020&nrg_bal=FC_IND_E&siec=E7000&unit=GWH	Stored as percent-number to match ECNO duplicate industry-electrification.
esabcc-i3-circular-mat-use	**updated**	**2022 = 11.4, 2023 = 12.1, 2024 = 12.2**	Eurostat cei_srm030 · circular material use rate · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/cei_srm030?format=JSON&geo=EU27_2020&unit=PC	Stored as percent-number — the TS series was rescaled ×100 (WP6 unit fix); the old ÷100 recipe failed its anchor check ever since. DB rows realigned by migration 075.
esabcc-f-gerd	**updated**	**2023 = 2.25, 2024 = 2.26**	Eurostat rd_e_gerdtot · GERD as % of GDP · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/rd_e_gerdtot?format=JSON&geo=EU27_2020&sectperf=TOTAL&unit=PC_GDP	Stored as percent of GDP. Publisher may revise the report's base year (vintage).
esabcc-o3-gross-inland	**updated**	**2022 = 15751.6, 2023 = 15111.2, 2024 = 15023.9**	Eurostat nrg_bal_c · gross inland consumption, all products · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_c?format=JSON&geo=EU27_2020&unit=GWH&nrg_bal=GIC&siec=TOTAL	GWh→TWh (÷1000). Direct — reproduces the report anchor 2021 at 1.000×.
esabcc-i5-industry-fec	**updated**	**2022 = 2638.9, 2023 = 2501.3, 2024 = 2506.3**	Eurostat nrg_bal_c · industry final energy consumption (energy use) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_c?format=JSON&geo=EU27_2020&unit=GWH&nrg_bal=FC_IND_E&siec=TOTAL	GWh→TWh (÷1000). Direct (anchor 2021 1.010×). FC_IND_E excludes non-energy use, matching the report definition.
esabcc-b2-buildings-fec	**updated**	**2022 = 4512.2, 2023 = 4306.2, 2024 = 4326.8**	Eurostat nrg_bal_c · residential (FC_OTH_HH_E) + services (FC_OTH_CP_E) final energy · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_c?format=JSON&geo=EU27_2020&unit=GWH&nrg_bal=FC_OTH_HH_E&siec=TOTAL	Spliced: the report's buildings boundary is ~7% wider than households+services alone (anchor 2021 0.932×), so only the year-on-year change is applied to the report baseline.

Indicator id	Status	New points (year = value)	Source (click)	Note
esabcc-b5a-residential-fossil-share	**updated**	**2022 = 43.83, 2023 = 42.08, 2024 = 41.74**	Eurostat nrg_bal_c · fossil (gas + oil excl. bio + solid) ÷ total residential final energy · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_c?format=JSON&geo=EU27_2020&unit=GWH&nrg_bal=FC_OTH_HH_E	Fossil = natural gas (G3000) + oil excl. biofuels (O4000XBIO) + solid fossil (C0000X0350-0370). Reproduces the report series to within 0.1 pp (anchor 2021 1.001×); spliced to hold the report baseline exactly.
esabcc-b5b-tertiary-fossil-share	**updated**	**2022 = 34.45, 2023 = 33.33, 2024 = 32.85**	Eurostat nrg_bal_c · fossil (gas + oil excl. bio + solid) ÷ total services final energy · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_c?format=JSON&geo=EU27_2020&unit=GWH&nrg_bal=FC_OTH_CP_E	Same fossil definition as B5a. The report's tertiary boundary sits ~1.9 pp above FC_OTH_CP_E (anchor 2021 0.951×), so the trend is spliced onto the report baseline rather than taken as a level.
esabcc-i7a-steel-projects	**error**	fetch failed: Cannot find package 'jszip' imported from /home/runner/work/ESABCCMethodHub/ESABCCMethodHub/scripts/esabcc-indicators/refresh-from-sources.mjs	LeadIT Green Steel Tracker · project database (2026 Q1) · cumulative EU-27 count https://www.industrytransition.org/green-steel-tracker/	Cumulative count of EU-27 projects by announcement year, from the tracker's own XLSX export. Reproduces the report's 2023 count to within 8% (52 vs 48); spliced so the report baseline is unchanged.
esabcc-i2-cement-use	**updated**	**2022 = 163.8, 2023 = 150.8, 2024 = 148.1**	Cement Europe (formerly Cembureau), Key Facts & Figures · "Cement Production And Consumption EU 27 & Cement Europe 2000-2024" · CONSUMPTION EU27 dataset https://www.cementeurope.eu/about-us/key-facts-figures/	Direct — the chart's inline dataset IS the report's series, extended: it reproduces every year the report carries to the tonne (2005 = 232,290,000 t vs the report's 232.3 Mt; 2013 = 142.2; 2020 = 159.2; 2021 = 170.5). The 2024 value (148.1 Mt) is the figure Cement Europe states in prose in its 2025 Activity Report. Still no CSV/XLSX/API — the numbers are a Chart.js dataset in the page HTML — so the parse is anchored on the dataset label, not its position.
esabcc-i7b-cement-projects	**updated**	**2026 = 145**	Cement Europe, Map of Innovation Projects · EU-27 projects in the page's inline `markers` array https://www.cementeurope.eu/innovation-projects/	Snapshot, not a rebuilt series: the map carries no announcement date (its only year is the plant's Operational Date), so the count is stamped with the run year. The report's 62-in-2023 cannot be reproduced from the current map, and the step to this value is part growth and part re-curation.
esabcc-i7c-chemicals-projects	**updated**	**2026 = 215**	Cefic, Low-Carbon Technologies Projects Map · /wp-json/wp/v2/gips, projects with at least one EU-27 country tag https://cefic.org/solutions-explained/low-carbon-technologies-projects/	Snapshot, not a rebuilt series: the only date on a record is its website posting date, and those cluster in the twice-yearly batches Cefic republishes the map in, so they date the curation and not the announcement. Counted as projects carrying at least one EU-27 country tag (several are multi-country).

Indicator id	Status	New points (year = value)	Source (click)	Note
esabcc-f-climate-patents-share	**updated**	**2020 = 12.32, 2021 = 9.41, 2022 = 6.86, 2023 = 6.75**	OECD DSD_PAT_DEV@DF_PAT_DEV · environment-related ÷ all-technology patent families (size 2+), EU-27 summed from member states https://data-explorer.oecd.org/vis?df[ds]=dsDisseminate&df[id]=DSD_PAT_DEV%40DF_PAT_DEV	Spliced: the current vintage revises the report years upward (report 2019 = 11.94 reads 14.44 today — counts fill in as families publish), so only the year-on-year change is applied to the report baseline. The newest one-two years are still filling in and revise upward on later runs; the refresh regenerates them each time. Member-state sum verified identical to OECD's own EU27_2020 ENV_PAT aggregate (fractional counting), which has no TOT counterpart.
esabcc-a7-bioenergy-feedstock	**updated**	**2022 = 25.19, 2023 = 24.56, 2024 = 24.15, 2025 = 24.51**	OECD-FAO Agricultural Outlook · EU biofuel use, wheat + maize + other coarse grains + vegetable oil https://data-explorer.oecd.org/vis?df[ds]=dsDisseminate&df[id]=DSD_AGR%40DF_OUTLOOK_2026_2035	Spliced proxy for the JRC medium-term outlook figure, whose own data is now DataM-token-gated with no published annex (probed 2026-07-30). Same Aglink-Cosimo family; wheat+maize+other-coarse-grains+vegetable-oil biofuel use reproduces the report within ~2% over 2010-2019, with a 4-10% vintage gap on 2020-21 — trend only, YoY change × report baseline. Newest edition resolved from the SDMX dataflow catalogue on each run; projection years (the edition id's first year onward) are excluded, so the series ends at the outlook's estimate year.
esabcc-i2-steel-use	**error**	fetch failed: Cannot find package 'pdfjs-dist' imported from /home/runner/work/ESABCCMethodHub/ESABCCMethodHub/scripts/esabcc-indicators/refresh-from-sources.mjs	Eurofer, European Steel in Figures (newest edition) · apparent steel consumption, chart data labels https://www.eurofer.eu/publications/brochures-booklets-and-factsheets/european-steel-in-figures-2026	Direct — the chart's data labels are the published figures ('000 t → Mt), and the 2021 anchor reproduces the report at 1.009×. The series row is identified by matching the report's own 2016-2020 values, so an edition that reorders the page cannot silently swap in real consumption. Edition discovered from the publications listing each run.
esabcc-a2-dairy-ghg	**updated**	**2022 = 97.17, 2023 = 96.63, 2024 = 94.57**	Eurostat env_air_gge CRF 3.A.1 + 3.B.1 (all cattle) split on Eurostat apro_mt_lscatl herd composition; split calibrated to UNFCCC DI 2021 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF3A1	Dairy share of all-cattle GHG allocated by head × emission factor, the factor ratio (≈2.68) calibrated so 2021 reproduces the inventory's own dairy share (49.4%). Spliced onto the report 2021 baseline.
esabcc-a2-bovine-ghg	**updated**	**2022 = 98.98, 2023 = 97.63, 2024 = 96.82**	As A2 (dairy, GHG), taking the non-dairy residual https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF3B1	Non-dairy (beef) share of all-cattle GHG, same construction as the dairy series. Spliced onto the report 2021 baseline.
esabcc-a2-dairy-ghg-intensity	**updated**	**2021 = 0.5999, 2022 = 0.5952, 2023 = 0.5886, 2024 = 0.5728**	Dairy-cattle GHG (see A2 dairy) ÷ Eurostat apro_mk_farm raw milk available · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mk_farm?format=JSON&geo=EU27_2020&dairyprod=D1100A&milkitem=PRO	Spliced ratio: Mt CO■eq ÷ thousand t cancels in the year-on-year ratio, so only the intensity trend is applied to the report baseline.

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esabcc-a2-bovine-ghg-intensity	**updated**	**2021 = 14.487, 2022 = 14.645, 2023 = 15.022, 2024 = 14.447**	Non-dairy cattle GHG (see A2 bovine) ÷ Eurostat apro_mt_pann bovine meat (B1000) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mt_pann?format=JSON&geo=EU27_2020&meat=B1000&meatitem=SLAUGHT&unit=THS_T	Spliced ratio, same construction as the dairy intensity.
esabcc-a4-bovine-consumption	**error**	fetch failed: Cannot find package 'jszip' imported from /home/runner/work/ESABCCMethodHub/ESABCCMethodHub/scripts/esabcc-indicators/refresh-from-sources.mjs	FAOSTAT Food Balance Sheets · bovine meat, food use, EU-27 sum https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_Europe.zip	EU-27 summed from member states (FAOSTAT publishes no EU aggregate; Cyprus comes from the Asia file). 1000 t → Mt. Spliced (anchor 2020 1.029×).
esabcc-a4-dairy-consumption	**error**	fetch failed: Cannot find package 'jszip' imported from /home/runner/work/ESABCCMethodHub/ESABCCMethodHub/scripts/esabcc-indicators/refresh-from-sources.mjs	FAOSTAT Food Balance Sheets · milk excl. butter, food use, EU-27 sum https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_Europe.zip	As A4 bovine. The report's dairy boundary is narrower than FBS "Milk – Excluding Butter" (anchor 2020 1.149×), so the splice applies the trend only.
esabcc-a4-pig-consumption	**error**	fetch failed: Cannot find package 'jszip' imported from /home/runner/work/ESABCCMethodHub/ESABCCMethodHub/scripts/esabcc-indicators/refresh-from-sources.mjs	FAOSTAT Food Balance Sheets · pigmeat, food use, EU-27 sum https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_Europe.zip	As A4 bovine. 1000 t → Mt. Spliced (anchor 2020 1.017×).
esabcc-a5-bovine-consumption	**error**	fetch failed: Cannot find package 'jszip' imported from /home/runner/work/ESABCCMethodHub/ESABCCMethodHub/scripts/esabcc-indicators/refresh-from-sources.mjs	FAOSTAT Food Balance Sheets · bovine meat, kcal/capita/day, EU-27 https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_Europe.zip	Total EU-27 kcal ÷ total EU-27 population ÷ 365 — computed on the aggregate, not as an unweighted mean of national per-capita figures. Spliced (anchor 2020 1.119×).
esabcc-a5-dairy-consumption	**error**	fetch failed: Cannot find package 'jszip' imported from /home/runner/work/ESABCCMethodHub/ESABCCMethodHub/scripts/esabcc-indicators/refresh-from-sources.mjs	FAOSTAT Food Balance Sheets · milk excl. butter, kcal/capita/day, EU-27 https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_Europe.zip	As A5 bovine (anchor 2021 0.765× — the report's dairy calorie boundary is wider than the FBS milk item, so only the trend is used).
esabcc-a5-pig-consumption	**error**	fetch failed: Cannot find package 'jszip' imported from /home/runner/work/ESABCCMethodHub/ESABCCMethodHub/scripts/esabcc-indicators/refresh-from-sources.mjs	FAOSTAT Food Balance Sheets · pigmeat, kcal/capita/day, EU-27 https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_Europe.zip	As A5 bovine (anchor 2020 0.920×).
esabcc-t6b-foodcrop-biofuels	**updated**	**2021 = 117.4, 2022 = 105.8, 2023 = 104.7, 2024 = 102.9**	Eurostat nrg_ind_urtd · biofuels from food and feed crops (R5280S), transport, RED basis · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_ind_urtd?format=JSON&geo=EU27_2020&nrg_bal=FC_TRA_E_RED&siec=R5280S&unit=KTOE	ktoe × 0.01163 → TWh. Direct: the RED-II series begins in 2021 so it never meets the 2015 report anchor, but the levels agree (2015 report 118.1 vs 2021 source 117.4).
esabcc-i4-steel-ghg-intensity	**updated**	**2022 = 0.9662, 2023 = 0.9413, 2024 = 0.911**	Eurostat env_air_gge CRF 2.C.1 iron & steel process GHG ÷ sts_inpr_a NACE C241 production index · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF2C1	Spliced ratio: process-emission intensity trend (2024 ≈ 0.947× the 2021 level) applied to the report baseline. Level is not sourced — Mt ÷ index cancels in the year-on-year ratio.
esabcc-i4-cement-ghg-intensity	**updated**	**2022 = 0.5875, 2023 = 0.5797, 2024 = 0.5614**	Eurostat env_air_gge CRF 2.A.1 cement-production process GHG ÷ sts_inpr_a NACE C235 production index · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF2A1	Spliced ratio: clinker-process intensity trend (2024 ≈ 0.930× the 2021 level) applied to the report baseline. Same C235 denominator as the I2 cement-production recipe.

Indicator id	Status	New points (year = value)	Source (click)	Note
esabcc-t3b-air-passenger	**error**	fetch failed: all sum legs failed	Eurostat avia_paoc · passengers carried on national + intra-EU27 flights · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/avia_paoc?format=JSON&geo=EU27_2020&unit=PAS&tra_meas=PAS_CRD&tra_cov=INTL_IEU27_2020&schedule=TOT	Passenger-km proxied by passengers carried: the pocketbook's published 2022/2023 Gpkm are kept and 2024/2025 derived from the year-on-year change in intra-EU passengers (constant average stage length). Validated on the overlap — passenger growth 2022→2023 (+14.7%) matches the pocketbook's Gpkm growth (+13.7%) to ~1 pp. Double counting of intra-EU legs is constant and cancels in the ratio.
esabcc-e1-energy-supply-ghg	**updated**	**2023 = 746.3, 2024 = 689.1**	Eurostat env_air_gge · CRF 1.A.1 + 1.B (energy supply) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF1A1	Public power/heat + fugitive emissions; CO■-eq (Mt).
esabcc-i1-industry-ghg	**updated**	**2023 = 622.1, 2024 = 611.4**	Eurostat env_air_gge · CRF 1.A.2 + 2 (industry energy + processes) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF1A2	CO■-eq (Mt).
esabcc-b1-buildings-ghg	**updated**	**2023 = 437.3, 2024 = 430.7**	Eurostat env_air_gge · CRF 1.A.4 (residential + commercial) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF1A4	CO■-eq (Mt).
esabcc-t1-transport-ghg	**updated**	**2023 = 901.9, 2024 = 920**	Eurostat env_air_gge · CRF 1.A.3 + intl aviation/navigation (1.D.1.a/b) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF1A3	Spliced: report baseline × env_air_gge transport YoY change (report scope includes international transport).
esabcc-a1-agri-nonco2	**updated**	**2023 = 359.9, 2024 = 355.7**	Eurostat env_air_gge · CRF 3 CH■+N■O (GWP-weighted) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=CH4&src_crf=CRF3	Spliced: GWP-weighted (CH■×28 + N■O×265) YoY change × report CO■-eq baseline.
esabcc-e6-energy-ch4	**updated**	**2022 = 64.46, 2023 = 60.05, 2024 = 57.77**	Eurostat env_air_gge · CRF 1 energy CH■ · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=CH4&src_crf=CRF1	Spliced: energy-sector CH■ mass YoY change × report CO■-eq baseline.
esabcc-l1-lulucf-net	**updated**	**2023 = -215.5, 2024 = -231**	Eurostat env_air_gge · CRF 4 (LULUCF) net · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF4	Net sink is negative. Subject to inventory-vintage revision vs the report base year.
esabcc-o1-ghg-total	**updated**	**2023 = 3118, 2024 = 3043.2**	Eurostat env_air_gge · net total incl. LULUCF (TOTXMEMO) + international aviation (1.D.1.a) + international navigation (1.D.1.b) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=TOTXMEMO	Direct sum (no splice), reproduces the report 2005-2022 within 0.1-1.5%.

Indicator id	Status	New points (year = value)	Source (click)	Note
esabcc-e2-fossil-power-share	**updated**	**2023 = 31.42, 2024 = 28.11**	Eurostat nrg_bal_peh · fossil ÷ total gross electricity production · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_peh?format=JSON&geo=EU27_2020&unit=GWH&nrg_bal=GEP	Spliced ratio: (solid fossil + peat + oil shale + oil + gas) ÷ TOTAL, YoY change × report baseline.
esabcc-e2-res-noBio-power-share	**updated**	**2023 = 40.4, 2024 = 43.21**	Eurostat nrg_bal_peh · (hydro + geothermal + wind + solar thermal + solar PV + tide) ÷ total gross electricity production · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_peh?format=JSON&geo=EU27_2020&unit=GWH&nrg_bal=GEP	Spliced ratio: non-biomass renewables ÷ TOTAL, YoY change × report baseline.
esabcc-e3-grid-co2-intensity	**updated**	**2023 = 208.7, 2024 = 187.4**	Eurostat env_air_gge CRF 1.A.1 ÷ nrg_bal_peh gross electricity production · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF1A1	Spliced trend proxy for the EEA intensity indicator; hand-check against the EEA-published value when available.
esabcc-t3a-road-share-passenger	**updated**	**2021 = 86.23, 2022 = 84.12, 2023 = 83.67, 2024 = 83.3**	Eurostat tran_hv_psmod · car + bus/coach share of inland passenger transport (CAR + BUS_TOT) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/tran_hv_psmod?format=JSON&geo=EU27_2020&unit=PC&vehicle=CAR	Spliced: Eurostat renamed/split the combined CAR_BUS_TOT vehicle code into separate CAR and BUS_TOT codes (2026-07); recipe now sums both legs. Eurostat modal-split scope (car+bus vs report car+two-wheeler) differs slightly; YoY change × report baseline.
esabcc-t5b-zev-lorries-stock	**updated**	**2023 = 6513, 2024 = 9831**	Eurostat road_eqs_lormot · battery-electric lorries >3.5t in the fleet (LOR_GT3P5) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/road_eqs_lormot?format=JSON&geo=EU27_2020&unit=NR&vehicle=LOR_GT3P5&mot_nrg=ELC	Spliced: Eurostat vehicle code LOR_HVY no longer exists; corrected code LOR_GT3P5 is on a narrower/different scope than the report's zero-emission-lorries figure (2022 Eurostat level 5425 vs report baseline 3794, 1.43x) — trend only, YoY change × report baseline.
esabcc-t6a-fossil-transport-share	**updated**	**2022 = 93.41, 2023 = 92.91, 2024 = 93.17**	Eurostat nrg_bal_s · fossil ÷ total transport final energy · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_s?format=JSON&geo=EU27_2020&unit=GWH&nrg_bal=FC_TRA_E	Spliced ratio: report scope includes intl bunkers, FC_TRA does not; YoY change × report baseline.
esabcc-l6-forest-sink	**updated**	**2022 = 236.7, 2023 = 255.7, 2024 = 258.3**	Eurostat env_air_gge · CRF 4.A forest land · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF4A	Spliced: report stores the sink as a positive magnitude (living biomass only) while CRF 4.A is negative net — the ratio preserves the report convention.
esabcc-l7-nonforest-lulucf	**updated**	**2022 = 121, 2023 = 135.6, 2024 = 123**	Eurostat env_air_gge · cropland + grassland + wetlands + settlements + other land (CRF 4.B–4.F) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF4B	Direct sum (no splice) of all five non-forest land categories; reproduces the report 2022-2024 figures directly (migration 078).
esabcc-l8-bioenergy-use	**updated**	**2022 = 1811.9, 2023 = 1766.3, 2024 = 1761.2**	Eurostat nrg_bal_s · gross inland consumption of biofuels + renewable waste · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_bal_s?format=JSON&geo=EU27_2020&unit=GWH&nrg_bal=GIC	Spliced sum (GWh→TWh); a failed siec leg is skipped, the splice ratio absorbs the level offset.

Indicator id	Status	New points (year = value)	Source (click)	Note
esabcc-a6-food-waste	**updated**	**2021 = 129, 2022 = 129, 2023 = 128, 2024 = 129**	Eurostat env_wasfw · total food waste, kg per capita · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_wasfw?format=JSON&geo=EU27_2020&nace_r2=TOT&unit=KG_HAB	Total food waste per capita (all NACE + households). Same unit as report; direct map.
esabcc-b4-population	**updated**	**2021 = 1.027, 2022 = 1.03, 2023 = 1.034, 2024 = 1.038, 2025 = 1.04**	Eurostat demo_gind · average population (indexed onto report 2005=1.0 baseline) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/demo_gind?format=JSON&geo=EU27_2020&indic_de=AVG	Spliced: absolute EU27 average population YoY change × the report's population index baseline.
esabcc-i2-steel-production	**updated**	**2022 = 136.3, 2023 = 125.9, 2024 = 128.8, 2025 = 128.8**	Eurostat sts_inpr_a · production-volume index (2021=100), NACE C241 basic iron & steel · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/sts_inpr_a?format=JSON&geo=EU27_2020&indic_bt=PRD&s_adj=CA&unit=I21&nace_r2=C241	Spliced: steel production-volume index YoY change × report 2021 baseline (152.8 Mt).
esabcc-i2-cement-production	**updated**	**2022 = 174.7, 2023 = 160.8, 2024 = 159.1, 2025 = 157.7**	Eurostat sts_inpr_a · production-volume index (2021=100), NACE C235 cement/lime/plaster · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/sts_inpr_a?format=JSON&geo=EU27_2020&indic_bt=PRD&s_adj=CA&unit=I21&nace_r2=C235	Spliced: cement-dominated NACE C235 production-volume index (cement-only C2351 has no EU27 series) YoY change × report 2021 baseline (182.5 Mt).
esabcc-i2-chemicals-production	**updated**	**2023 = 23.07, 2024 = 24.75, 2025 = 20.01**	Eurostat PRODCOM ds-059358/ds-059367 · sold production of the report's seven base organic chemicals · EU27_2020 https://ec.europa.eu/eurostat/api/comext/dissemination/sdmx/3.0/data/dataflow/ESTAT/ds-059358/1.0?c[reporter]=EU27_2020&c[product]=20141130,20141140,20141223,20141225,20141243,20141245,20141247&c[indicators]=PRODQNT	The report's own method, re-run: sum of sold production (PRODQNT) of ethylene, propylene, benzene, toluene and the three xylenes. Absolute tonnage, not a spliced index.
esabcc-i2-chemicals-use	**updated**	**2023 = 24.31, 2024 = 26.2, 2025 = 21.37**	Eurostat PRODCOM ds-059358/ds-059367 · sold production + imports – exports of the report's seven base organic chemicals · EU27_2020 https://ec.europa.eu/eurostat/api/comext/dissemination/sdmx/3.0/data/dataflow/ESTAT/ds-059358/1.0?c[reporter]=EU27_2020&c[product]=20141130,20141140,20141223,20141225,20141243,20141245,20141247&c[indicators]=PRODQNT,IMPQNT,EXPQNT	Apparent use = production + imports – exports, over the same seven products the report used.
esabcc-i2-chemicals-trade-balance	**updated**	**2023 = -1.239, 2024 = -1.457, 2025 = -1.357**	Eurostat PRODCOM ds-059358/ds-059367 · exports – imports of the report's seven base organic chemicals · EU27_2020 https://ec.europa.eu/eurostat/api/comext/dissemination/sdmx/3.0/data/dataflow/ESTAT/ds-059358/1.0?c[reporter]=EU27_2020&c[product]=20141130,20141140,20141223,20141225,20141243,20141245,20141247&c[indicators]=IMPQNT,EXPQNT	Trade balance = exports – imports (negative = net importer), the report's own sign convention.
esabcc-i2-steel-trade-balance	**updated**	**2022 = -26.955, 2023 = -23.734, 2024 = -23.811, 2025 = -30.869**	Eurostat ds-059366 · extra-EU27 exports – imports of CPA 2410 basic iron, steel & ferro-alloys, quantity · EU27_2020 https://ec.europa.eu/eurostat/api/comext/dissemination/sdmx/3.0/data/dataflow/ESTAT/ds-059366/1.0?c[reporter]=EU27_2020&c[partner]=EXT_EU27_2020&c[product]=2410&c[indicators]=QUANTITY_KG	Reproduces the report's 2008-2021 series to 5e-5 relative error, so taken directly rather than spliced.

Indicator id	Status	New points (year = value)	Source (click)	Note
esabcc-a2-bovine-production	**updated**	**2021 = 6.802, 2022 = 6.64, 2023 = 6.385, 2024 = 6.584, 2025 = 6.32**	Eurostat apro_mt_pann · bovine meat (B1000) slaughterings · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mt_pann?format=JSON&geo=EU27_2020&meat=B1000&meatitem=SLAUGHT&unit=THS_T	Slaughterings = production; thousand t → Mt (÷1000). Direct (anchor 2020 0.984×).
esabcc-a4-bovine-production	**updated**	**2021 = 6.802, 2022 = 6.64, 2023 = 6.385, 2024 = 6.584, 2025 = 6.32**	Eurostat apro_mt_pann · bovine meat (B1000) slaughterings · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mt_pann?format=JSON&geo=EU27_2020&meat=B1000&meatitem=SLAUGHT&unit=THS_T	Same series as esabcc-a2-bovine-production.
esabcc-a2-pig-production	**updated**	**2021 = 23.39, 2022 = 22.07, 2023 = 20.64, 2024 = 21.09, 2025 = 21.84**	Eurostat apro_mt_pann · pigmeat (B3100) slaughterings · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mt_pann?format=JSON&geo=EU27_2020&meat=B3100&meatitem=SLAUGHT&unit=THS_T	Pig code B3100 (not B4000 = sheep/goats); thousand t → Mt. Direct (anchor 2020 0.992×).
esabcc-a4-pig-production	**updated**	**2021 = 23.39, 2022 = 22.07, 2023 = 20.64, 2024 = 21.09, 2025 = 21.84**	Eurostat apro_mt_pann · pigmeat (B3100) slaughterings · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mt_pann?format=JSON&geo=EU27_2020&meat=B3100&meatitem=SLAUGHT&unit=THS_T	Same series as esabcc-a2-pig-production.
esabcc-a2-dairy-production	**updated**	**2021 = 162.6, 2022 = 162.8, 2023 = 163.7, 2024 = 164.6**	Eurostat apro_mk_farm · raw milk total available on farms · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mk_farm?format=JSON&geo=EU27_2020&dairyprod=D1100A&milkitem=PRO	Spliced: raw-milk-available scope differs slightly from the report dairy-production boundary (anchor 2020 0.983×); 1000 t → Mt.
esabcc-a4-dairy-production	**updated**	**2021 = 162.6, 2022 = 162.8, 2023 = 163.7, 2024 = 164.6**	Eurostat apro_mk_farm · raw milk total available on farms · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mk_farm?format=JSON&geo=EU27_2020&dairyprod=D1100A&milkitem=PRO	Same series as esabcc-a2-dairy-production.
esabcc-a4-bovine-herd-size	**updated**	**2021 = 59.5, 2022 = 58.79, 2023 = 57.97, 2024 = 56.5, 2025 = 56.26**	Eurostat apro_mt_lscatl · live bovine animals (A2000), Nov–Dec survey · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mt_lscatl?format=JSON&geo=EU27_2020&animals=A2000&month=M11_M12&unit=THS_HD	Spliced: A2000 total cattle includes dairy, report bovine-meat herd is a subset (anchor 1.272×); thousand head → million head.
esabcc-a4-dairy-herd-size	**updated**	**2021 = 20.21, 2022 = 20.07, 2023 = 19.91, 2024 = 19.22, 2025 = 19.02**	Eurostat apro_mt_lscatl · dairy cows (A2300F), Nov–Dec survey · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mt_lscatl?format=JSON&geo=EU27_2020&animals=A2300F&month=M11_M12&unit=THS_HD	Dairy cows; thousand head → million head. Direct (anchor 2020 0.995×).
esabcc-a4-pig-herd-size	**updated**	**2021 = 135.9, 2022 = 129, 2023 = 127.5, 2024 = 126.8, 2025 = 126.1**	Eurostat apro_mt_lspig · live swine (A3100), Nov–Dec survey · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_mt_lspig?format=JSON&geo=EU27_2020&animals=A3100&month=M11_M12&unit=THS_HD	Pig herd code A3100 (not A5000); thousand head → million head. Spliced onto report baseline (anchor 1.042×).
esabcc-a2-pig-ghg	**updated**	**2022 = 28.01, 2023 = 26.82, 2024 = 26.18**	Eurostat env_air_gge · swine enteric (CRF 3.A.3) + manure (CRF 3.B.3) GHG · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF3A3	Spliced: on-farm swine GHG (CO■-eq) YoY change × report baseline (anchor 2021 0.929×). Cattle cannot be split into bovine-meat vs dairy in the inventory.
esabcc-a2-pig-ghg-intensity	**updated**	**2021 = 1.219, 2022 = 1.266, 2023 = 1.296, 2024 = 1.238**	Eurostat env_air_gge swine GHG (3.A.3 + 3.B.3) ÷ apro_mt_pann pigmeat (B3100) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/env_air_gge?format=JSON&geo=EU27_2020&unit=MIO_T&airpol=GHG&src_crf=CRF3B3	Spliced ratio: swine GHG ÷ pig production; num/den unit mix cancels in the YoY ratio, only the trend is applied to the report intensity baseline (conceptual 2020 0.931×).

Indicator id	Status	New points (year = value)	Source (click)	Note
esabcc-e4a-solar-pv-add	**updated**	**2022 = 33.34, 2023 = 51.63, 2024 = 60.96**	Eurostat nrg_inf_epcrw · net electrical capacity, solar photovoltaic (RA420) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_inf_epcrw?format=JSON&geo=EU27_2020&unit=MW&plant_tec=CAP_NET_ELC&siec=RA420	Year-on-year delta of net capacity stock (MW→GW), direct (no splice — reproduces the report anchor within a few %). Eurostat has no 2025 row yet (latest year 2024), so the series stops there; SolarPower Europe/WindEurope market-report figures are a different (gross-installation) basis and are not used.
esabcc-e4b-wind-add	**updated**	**2022 = 15.88, 2023 = 12.12, 2024 = 15.32**	Eurostat nrg_inf_epcrw · net electrical capacity, wind onshore+offshore (RA300) · EU27_2020 https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/nrg_inf_epcrw?format=JSON&geo=EU27_2020&unit=MW&plant_tec=CAP_NET_ELC&siec=RA300	Year-on-year delta of net capacity stock (MW→GW), direct (no splice — reproduces the report anchor within a few %). Eurostat has no 2025 row yet (latest year 2024), so the series stops there; WindEurope/SolarPower Europe market-report figures are a different (gross-installation) basis and are not used.

Indicators not covered by an automated recipe (transport demand, ZEV lorries, heat pumps, green-bond/cleantech finance, intensity ratios) remain in the curated sheet *Indicator-Database-Post-Report-Update.pdf*. Extend the RECIPES table in refresh-from-sources.mjs to bring more under automation.